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A MORE RELATED WORK

Task-specific Learning. We furnish an extension of the related work on task-specific learning, focusing particularly on the most related models to which we made comparisons. Recent works improve Transformer (Vaswani et al., 2017) for time series forecasting by incorporating signal processing principles like patching, exponential smoothing, decomposition, and frequency analysis. For example, PatchTST (Nie et al., 2023) segments time series into patches as input tokens to Transformer. This retains local semantics, reduces computation/memory for attention, and allows longer history. It improves long-term forecast accuracy over other Transformer models. It also achieves excellent performance on self-supervised pretraining and transfer learning. ETSformer (Woo et al., 2022) incorporates exponential smoothing principles into Transformer attention to improve accuracy and efficiency. It uses exponential smoothing attention and frequency attention to replace standard self-attention. FEDformer (Zhou et al., 2022) combines Transformer with seasonal-trend decomposition. The decomposition captures the global profile while Transformer captures detailed structures. It also uses frequency enhancement for long-term prediction. This provides better performance and efficiency than the standard Transformer. Autoformer (Wu et al., 2021) uses a decomposition architecture with auto-correlation to enable progressive decomposition capacities for complex series. Auto-correlation is designed based on series periodicity to conduct dependency discovery and representation aggregation. It outperforms self-attention in efficiency and accuracy.

Although these methods enhance efficiency and accuracy compared to vanilla Transformer, they are mostly designed and optimized for narrow prediction tasks within specific domains. These models are typically trained end-to-end on small, domain-specific datasets. While achieving strong performance on their target tasks, such specialized models sacrifice versatility and generalizability across the diverse range of time series data encountered in the real world. The narrow focus limits their applicability to new datasets and tasks. To advance time series forecasting, there is a need for more flexible, widely applicable models that can adapt to new data distributions and tasks without extensive retraining. Ideal models would learn robust time series representations that transfer knowledge across domains. Developing such broadly capable forecasting models remains an open challenge. According to our discussions of related previous work, recent studies have begun to explore model versatility through pre-training and architectural innovations. However, further efforts are needed to realize the truly general-purpose forecasting systems that we are advancing in this research.

Cross-modality Adaptation. We provide an extended overview of related work in cross-modality adaptation, with a particular focus on recent advancements in model reprogramming for time series and other data modalities. Model reprogramming is a resource-efficient cross-domain learning approach that involves adapting a well-developed, pre-trained model from one domain (source) to address tasks in a different domain (target) without the need for model fine-tuning, even when these domains are significantly distinct, as noted by Chen (2022). In the context of time series data, Voice2Series (Yang et al., 2021) adapts an acoustic model from speech recognition for time series classification by transforming the time series to fit the model and remapping outputs to new labels. Similarly, LLMTime (Gruver et al., 2023) adapts LLMs for zero-shot time series forecasting, focusing on the effective tokenization of input time series for the backbone LLM, which then generates forecasts autoregressively. Diverging from these methods, TIME-LLM does not edit the input time series directly. Instead, it proposes reprogramming time series with the source data modality along with prompting to unleash the full potential of LLMs as versatile forecasters in standard, few-shot, and zero-shot scenarios. Other notable works in this field, mostly in biology, include R2DL (Vinod et al., 2020) and ReproBert (Melnyk et al., 2023), which reprogram amino acids using word embeddings. A key distinction with our patch reprogramming approach is that, unlike the complete set of amino acids, time series patches do not form a complete set. Thus, we propose optimizing a small set of text prototypes and their mapping to time series patches, rather than directly optimizing a large transformation matrix between two complete sets, such as vocabulary and amino acids.

B EXPERIMENTAL DETAILS

B.1 IMPLEMENTATION

We mainly follow the experimental configurations in (Wu et al., 2023) across all baselines within a unified evaluation pipeline in <https://github.com/thuml/Time-Series-Library> for

fair comparisons. We use Llama-7B (Touvron et al., 2023) as the default backbone model unless stated otherwise. All our experiments are repeated three times and we report the averaged results. Our model implementation is on PyTorch (Paszke et al., 2019) with all experiments conducted on NVIDIA A100-80G GPUs. Our detailed model configurations are in Appendix B.4, and our code is made available at <https://github.com/KimMeen/Time-LLM>.

Technical Details. We provide additional technical details of TIME-LLM in three aspects: (1) the learning of text prototypes, (2) the calculation of trends and lags in time series for use in prompts, and (3) the implementation of the output projection. To identify a small set of text prototypes $\mathbf{E}' \in \mathbb{R}^{V' \times D}$ from $\mathbf{E} \in \mathbb{R}^{V \times D}$, we learn a matrix $\mathbf{W} \in \mathbb{R}^{V' \times V}$ as the intermediary. To describe the overall time series trend in natural language, we calculate the sum of differences between consecutive time steps. A sum greater than 0 indicates an upward trend, while a lesser sum denotes a downward trend. In addition, we calculate the top-5 lags of the time series, identified by computing the autocorrelation using fast Fourier transformation and selecting the five lags with the highest correlation values. After we pack and feedforward the prompt and patch embeddings $\mathbf{O}^{(i)} \in \mathbb{R}^{P \times D}$ through the frozen LLM, we discard the prefixal part and obtain the output representations, denoted as $\tilde{\mathbf{O}}^i \in \mathbb{R}^{P \times D}$. Subsequently, we follow PatchTST (Nie et al., 2023) and flatten $\tilde{\mathbf{O}}^i$ into a 1D tensor with the length $P \times D$, which is then linear projected as $\hat{\mathbf{Y}}^i \in \mathbb{R}^H$.

B.2 DATASET DETAILS

Dataset statistics are summarized in Tab. 8. We evaluate the long-term forecasting performance on the well-established eight different benchmarks, including four ETT datasets (Zhou et al., 2021) (i.e., ETTh1, ETTh2, ETTm1, and ETTm2), Weather, Electricity, Traffic, and ILI from (Wu et al., 2023). Furthermore, we evaluate the performance of short-term forecasting on the M4 benchmark (Makridakis et al., 2018) and the quarterly dataset in the M3 benchmark (Makridakis & Hibon, 2000).

Table 8: Dataset statistics are from (Wu et al., 2023). The dimension indicates the number of time series (i.e., channels), and the dataset size is organized in (training, validation, testing).

Tasks	Dataset	Dim.	Series Length	Dataset Size	Frequency	Domain
Long-term Forecasting	ETTh1	7	{96, 192, 336, 720}	(34465, 11521, 11521)	15 min	Temperature
	ETTh2	7	{96, 192, 336, 720}	(34465, 11521, 11521)	15 min	Temperature
	ETTh1	7	{96, 192, 336, 720}	(8545, 2881, 2881)	1 hour	Temperature
	ETTh2	7	{96, 192, 336, 720}	(8545, 2881, 2881)	1 hour	Temperature
	Electricity	321	{96, 192, 336, 720}	(18317, 2633, 5261)	1 hour	Electricity
	Traffic	862	{96, 192, 336, 720}	(12185, 1757, 3509)	1 hour	Transportation
	Weather	21	{96, 192, 336, 720}	(36792, 5271, 10540)	10 min	Weather
	ILI	7	{24, 36, 48, 60}	(617, 74, 170)	1 week	Illness
Short-term Forecasting	M3-Quarterly	1	8	(756, 0, 756)	Quarterly	Multiple
	M4-Yearly	1	6	(23000, 0, 23000)	Yearly	Demographic
	M4-Quarterly	1	8	(24000, 0, 24000)	Quarterly	Finance
	M4-Monthly	1	18	(48000, 0, 48000)	Monthly	Industry
	M4-Weakly	1	13	(359, 0, 359)	Weakly	Macro
	M4-Daily	1	14	(4227, 0, 4227)	Daily	Micro
	M4-Hourly	1	48	(414, 0, 414)	Hourly	Other

The Electricity Transformer Temperature (ETT; An indicator reflective of long-term electric power deployment) benchmark is comprised of two years of data, sourced from two counties in China, and is subdivided into four distinct datasets, each with varying sampling rates: ETTh1 and ETTh2, which are sampled at a 1-hour level, and ETTm1 and ETTm2, which are sampled at a 15-minute level. Each entry within the ETT datasets includes six power load features and a target variable, termed “oil temperature”. The Electricity dataset comprises records of electricity consumption from 321 customers, measured at a 1-hour sampling rate. The Weather dataset includes one-year records